CH 5960: The Nuclear Energy Option

First Assignment : Monte Carlo Simulations of Reactor Physics

Through Monte Carlo simulations of neutron movement inside a homogeneous core, show that an infinite reactor

1. containing pure U-238 cannot go critical
2. containing only natural uranium cannot go critical
3. containing 2% enriched uranium and light water as moderator can go critical
4. containing natural uranium and heavy water as moderator can go critical

Assume that

* neutrons are born with kinetic energy as per the distribution
  s(E) = 0.771 (E^1/2) exp(-0.776E) where E is in MeV

* Use energy-dependent microscopic cross-sections, for example from the website https://atom.kaeri.re.kr/nuchart/?zlv=0#

Ensure that in your code you

* tabulate microscopic cross-sections (elastic, inelastic, capture and fission) for U-238 etc. at neutron energies, in MeV of 0.2, 0.4, 0.6, 0.8. 1.0x 10^y) where y = -8, -7, -6, -5, -4, -3, -2, -1, 0, 1 as a minimum resolution

* track (i) one million and (ii) one billion neutrons to get good statistics

In your results, you should bring sufficient number of graphs to show that your model results are conforming to expectations, e.g., of

* neutron birth energy distribution,
* neutron slowing down process
* neutron lifetimes as fast, slowing down and thermal neutrons,
* absorption in the moderator vs fuel,
* fission-to-capture ratio etc.

Deadline for submission of code and report: October 5th, 2026, 8.26 pm